# MSG No. 76.12 $P_I4_1$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1,8]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{4}\right]$ | `[2,7]` |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$ | `[3,6]` |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[4,5]` |

Table 2: Wyckoff site: `8b`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1]` |
| 2 | $\left[-y, x, z+\frac{1}{4}\right]$ | `[2]` |
| 3 | $\left[y, -x, z+\frac{3}{4}\right]$ | `[3]` |
| 4 | $\left[-x, -y, z+\frac{1}{2}\right]$ | `[4]` |
| 5 | $\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[\frac{1}{2}-y, x+\frac{1}{2}, z+\frac{3}{4}\right]$ | `[6]` |
| 7 | $\left[y+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{4}\right]$ | `[7]` |
| 8 | $\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$ | `[8]` |